

# Report: AI Agent Pipeline With Privacy Filtering & Moderation

## 1. Introduction

AI systems must prevent unsafe or sensitive user input from reaching an LLM. Users may provide personal information, profanity, or harmful intent.

To ensure safety, this project implements a simple AI Agent Pipeline containing:

- **Privacy Filter** – redacts PII
- **Moderation Filter** – removes profanity and adult content
- **LLM Safety Processor** – blocks harmful intent

This report summarizes the design and limitations of the system.

## 2. System Architecture

The pipeline follows a linear, LangGraph-style workflow:

**User Input → Privacy Filter → Moderation Filter → LLM Safety Processor → Output**

Each stage cleans or validates the input before passing it forward, ensuring only safe text reaches the LLM.

## 3. Design Decisions

### 3.1 Privacy Filter

- Implemented using regex patterns for emails, phone numbers, credit card numbers, and addresses.
- Regex was chosen for being simple, fast, and easy to maintain.
- Sensitive items are replaced with placeholders like: [EMAIL], [PHONE], [CARD], [ADDRESS].

This prevents accidental storage or misuse of PII by the LLM.

### 3.2 Moderation Filter

- Uses a small dictionary of profanities and slang.
- Regex patterns detect:
  - Normal profanity
  - Leetspeak (e.g., f@ck, sh1t)
  - Words with separators (f\*ck, fu#k)

- A normalization function converts symbols (e.g., @ → a, \$ → s) to improve detection.

This ensures that disguised offensive words are still filtered out.

### 3.3 LLM Safety Processor

- A keyword-based detector checks for:
  - Violence
  - Self-harm
  - Sexual content
  - Hate speech
  - Illegal activity
- If detected, the system blocks the prompt and returns an “Unsafe request” message.

This prevents the LLM from being misused for harmful purposes.

## 4. Workflow Summary

1. User enters text
2. Privacy Filter removes all detected PII
3. Moderation Filter censors profanity & explicit terms
4. Safety Processor checks intent
5. Cleaned and safe prompt is finally sent to the LLM

This ensures privacy + safe content + responsible responses.

## 5. Limitations

- **Regex-based PII detection** may miss unusual formats, non-English data, or heavily obfuscated text.
- **Small profanity list** :expanding it or using a toxicity model would increase accuracy.
- **Keyword-based safety detection**: lacks deep contextual understanding.
- **English-only** :other languages are not well supported.
- **No memory** : the system evaluates each input independently.

## 6. Conclusion

The pipeline demonstrates a simple but effective approach to privacy and safety in AI systems. Its modular design makes it easy to extend, while fast regex and dictionary-based methods keep it lightweight.

Although it has limitations, the system fulfills all core requirements for filtering PII, moderating content, and protecting the LLM from harmful prompts.